

# Module 4

## Circuit Analysis in the S Domain

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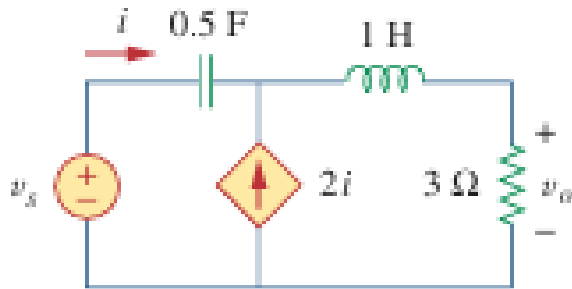
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# Tutorial

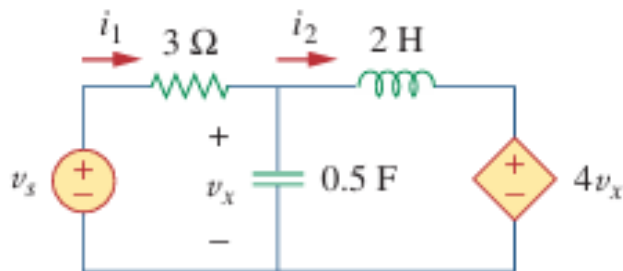
1. The transfer function of a certain circuit is given. Find the impulse response of the circuit.

$$H(s) = \frac{5}{s+1} - \frac{3}{s+2} + \frac{6}{s+4}$$

2. Obtain the transfer function  $H(s) = V_o / V_s$  for the circuit given below.



3. For the circuit given below, find: a)  $I_1 / V_s$     b)  $I_2 / V_x$

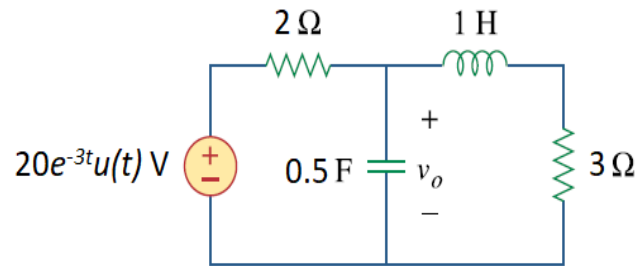


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4. The transfer function of a system is given.  $H(s) = \frac{s^2}{3s + 1}$

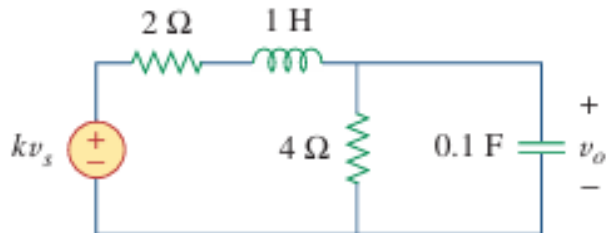
Find the output when the system has an input of  $4 e^{-t/3} u(t)$ .

5. Apply the Thevenin theorem to determine  $v_o(t)$  for  $t > 0$  for the circuit given in Fig.8



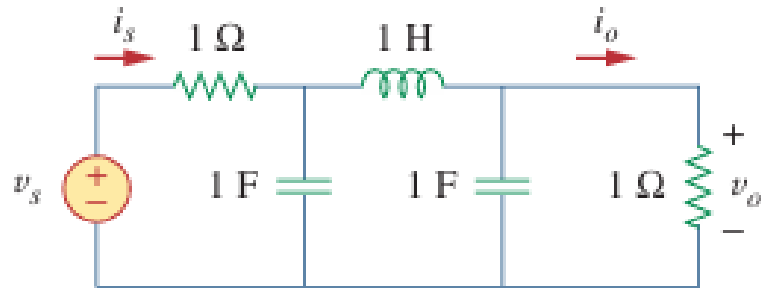
6. When the input to a system is a unit step function, the response is  $10 \cos 2t (t)$ . Obtain the transfer function of the system.

7. For the circuit below, find  $H(s) = V_o(s)/V_s(s)$ . Assume zero initial conditions.



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8. Refer to the network given below. Find the following transfer functions:



(a)  $H_1(s) = V_o(s)/V_s(s)$

(b)  $H_2(s) = V_o(s)/I_s(s)$

(c)  $H_3(s) = I_o(s)/I_s(s)$

(d)  $H_4(s) = I_o(s)/V_s(s)$